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Sakurada et al.

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(54) **ELECTRIC MOTOR**

(71) Applicant: **Mitsubishi Electric Corporation**, Tokyo (JP)

(72) Inventors: **Tetsuya Sakurada**, Tokyo (JP); **Kazuki Iwasa**, Tokyo (JP)

(73) Assignee: **Mitsubishi Electric Corporation**, Tokyo (JP)

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H02K 9/06 (2006.01)

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CPC ***H02K 1/32*** (2013.01); ***H02K 9/06*** (2013.01)

(58) **Field of Classification Search**
CPC H02K 1/32; H02K 5/207; H02K 9/06; H02K 5/15
See application file for complete search history.

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Primary Examiner — Terrance L Kenerly
Assistant Examiner — Theodore L Perkins
(74) *Attorney, Agent, or Firm* — BUCHANAN, INGERSOLL & ROONEY PC

(57) **ABSTRACT**

An electric motor includes a shaft supported rotatably around a rotation axis, a rotor rotatable integrally with the shaft, a stator, and first and second retaining members to hold the rotor therebetween in an extending direction of the rotation axis. The rotor includes a stack of thin plates arranged in the extending direction of the rotation axis and including identically-shaped through holes. The through holes define air passages penetrating the rotor in the extending direction of the rotation axis. The first and second retaining members respectively include first and second through holes in communication with the air passages. The radially outermost portions of the walls of the first through holes are located radially inward from the radially outermost portions of the walls of the second through holes.

9 Claims, 16 Drawing Sheets

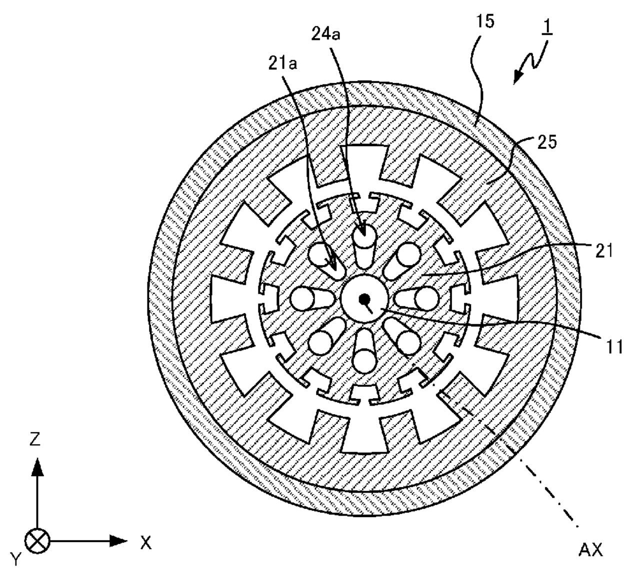

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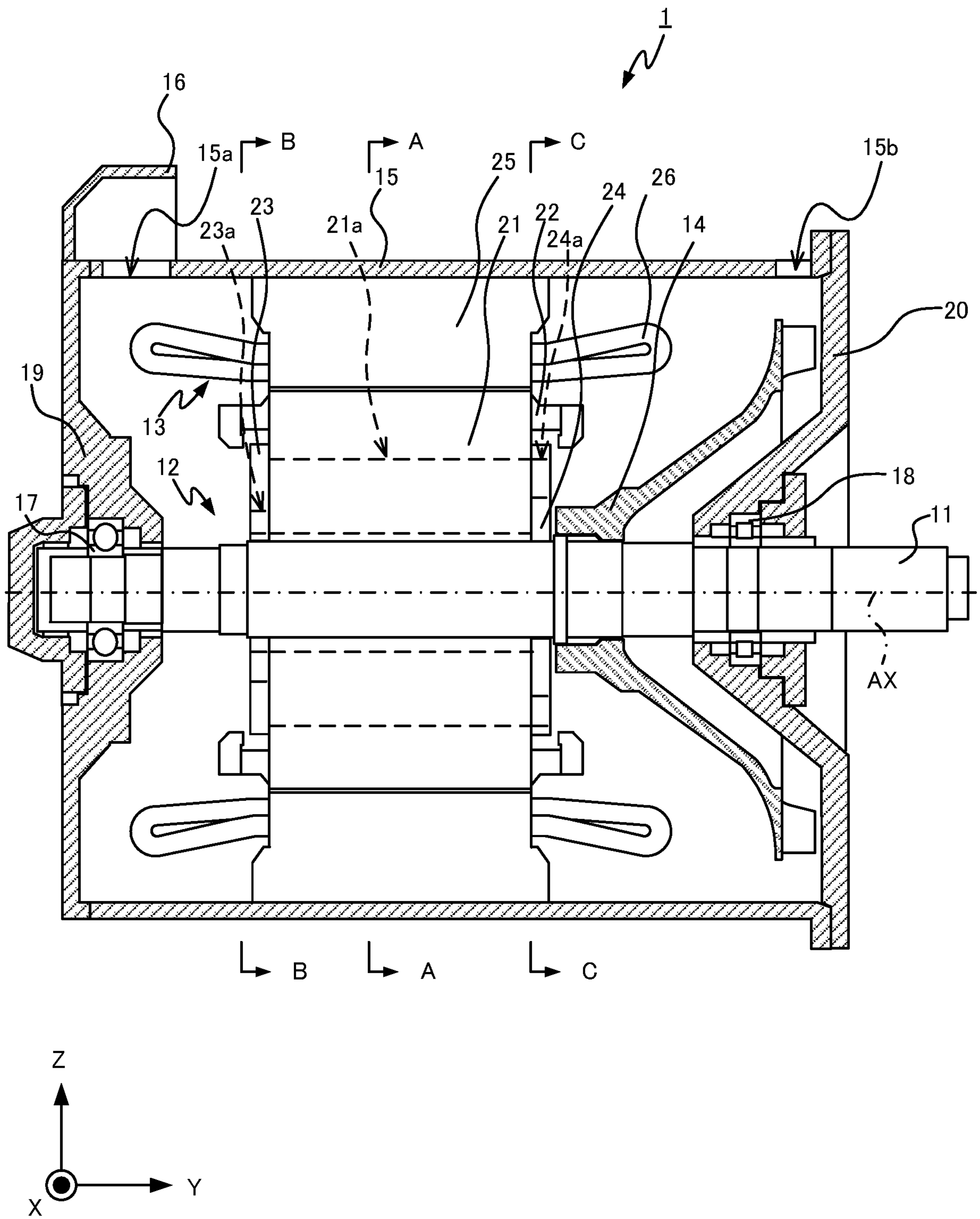
FIG.1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21a
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FIG.2

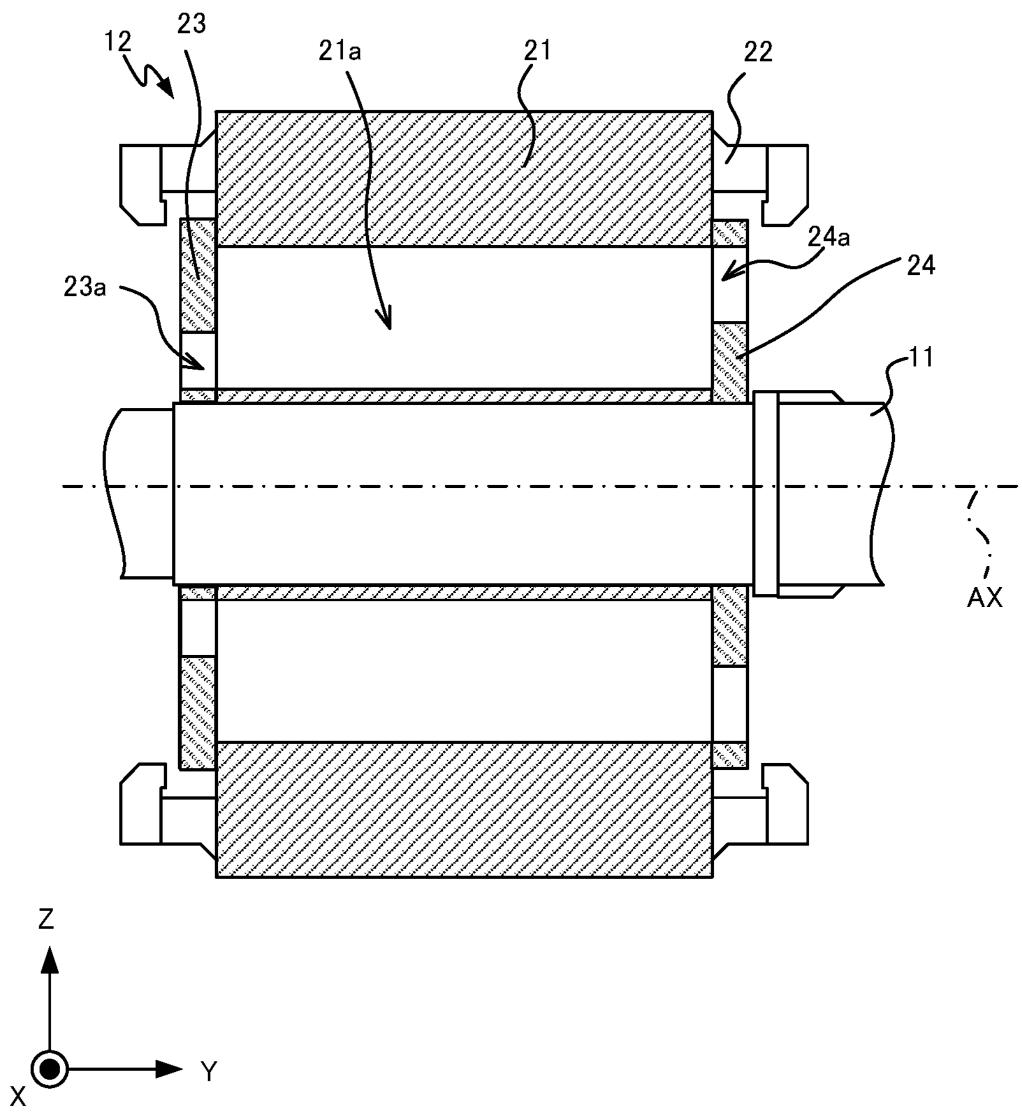


FIG.3

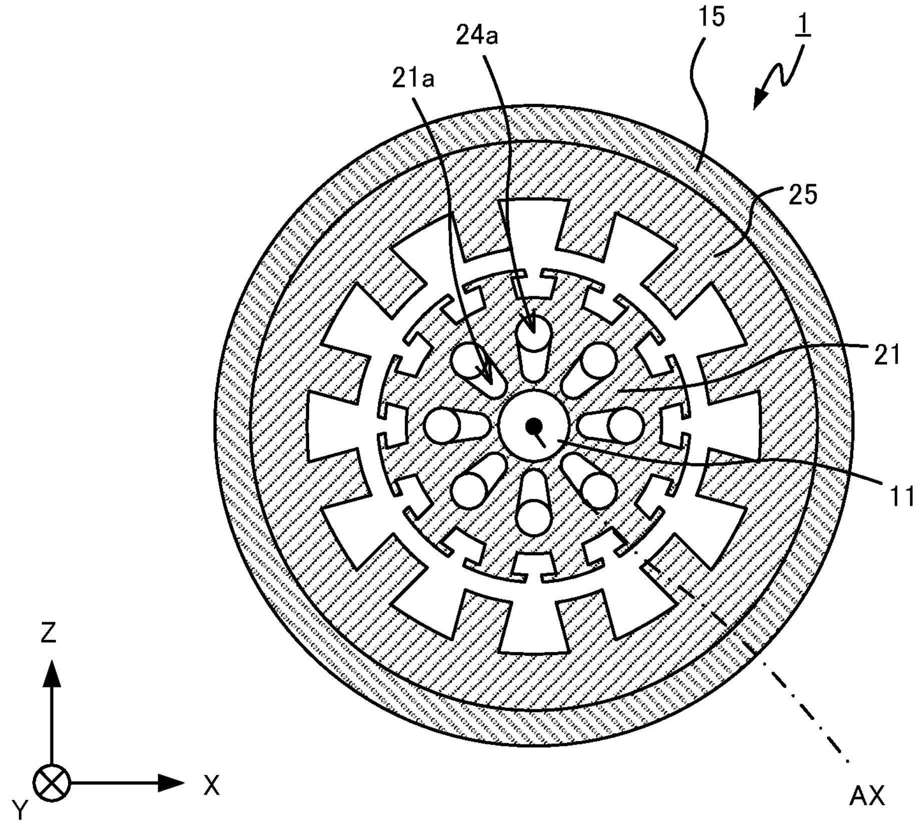


FIG.4

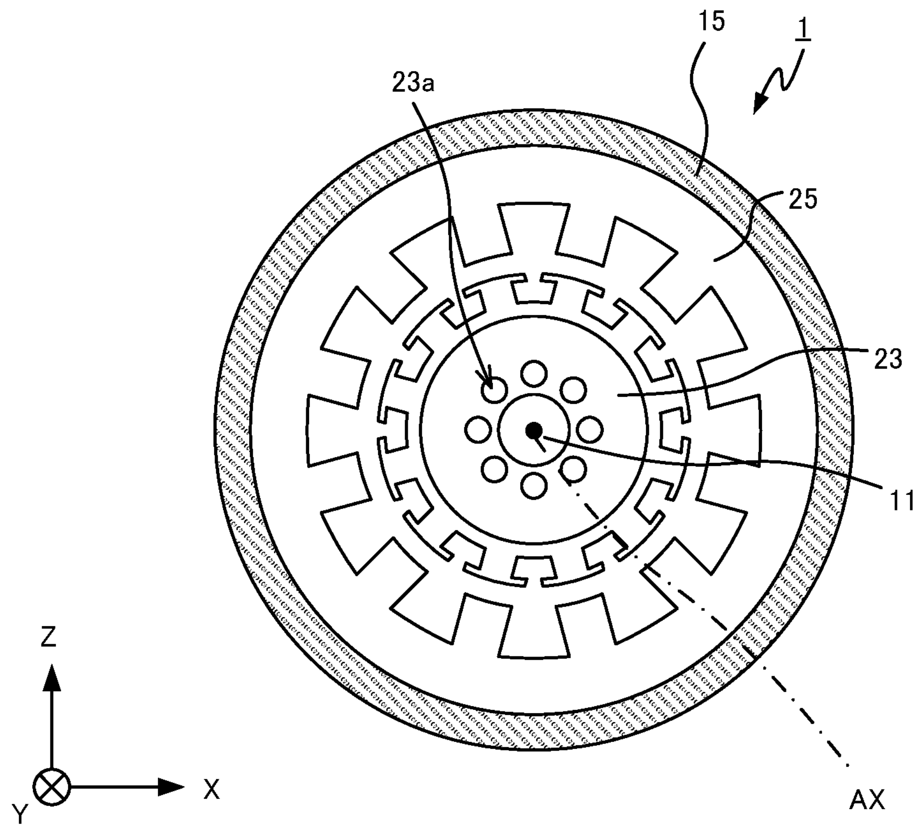


FIG.5

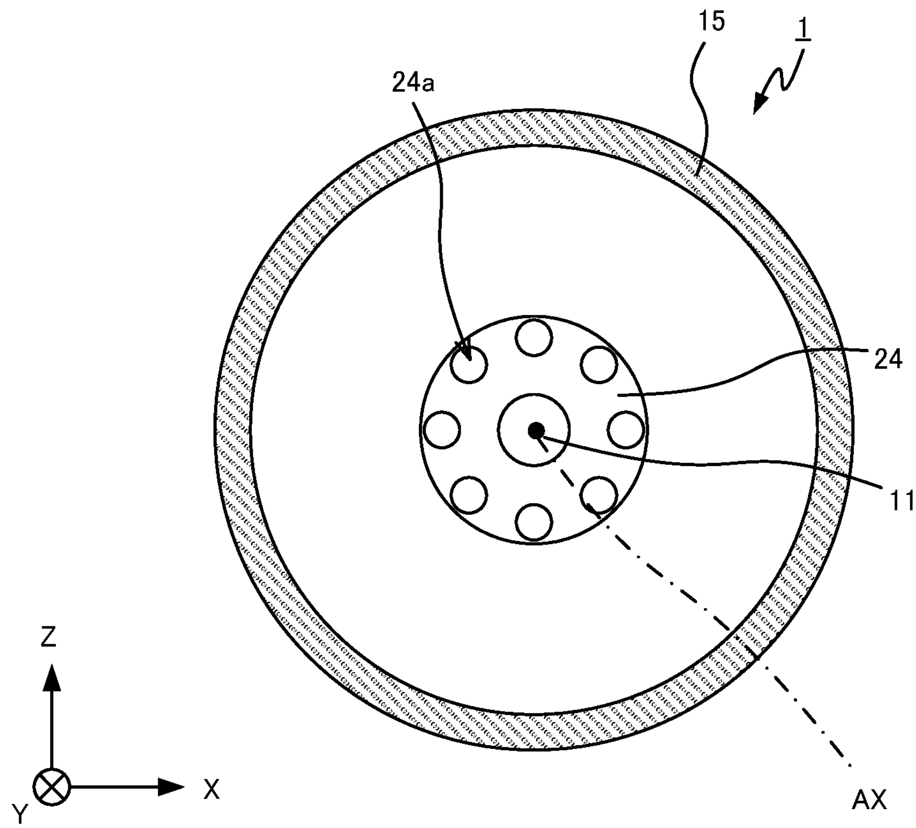


FIG.6

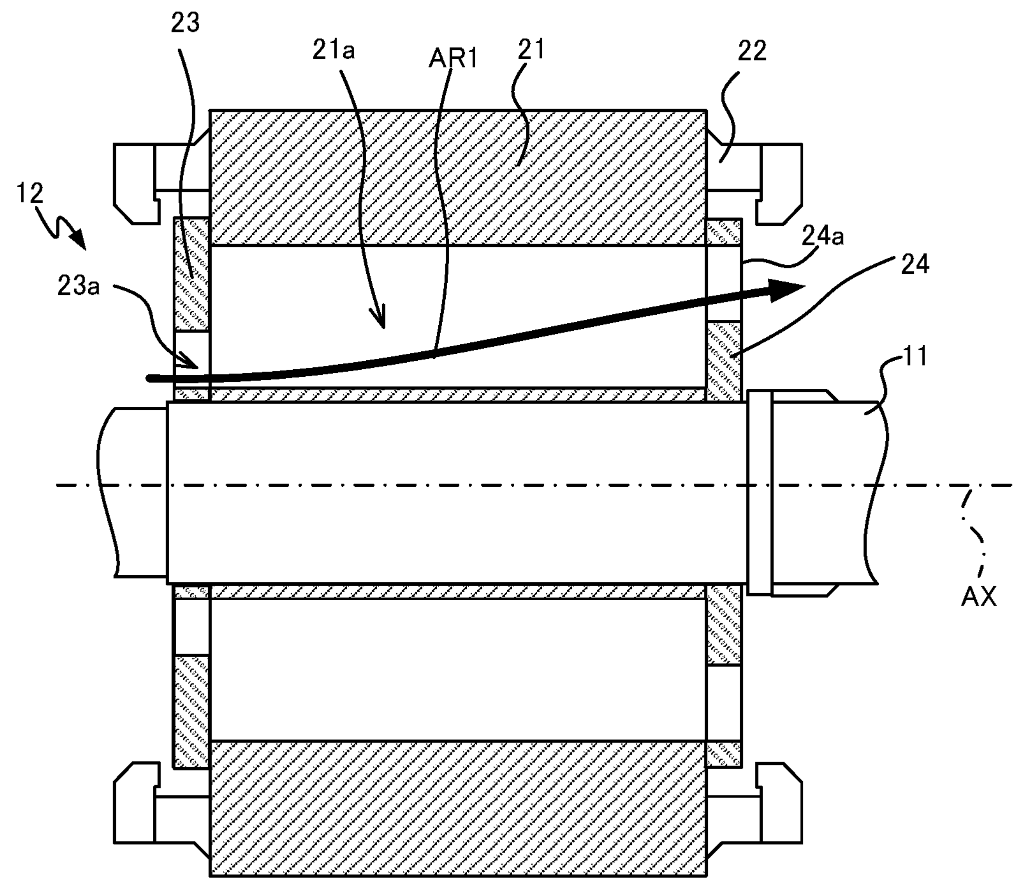
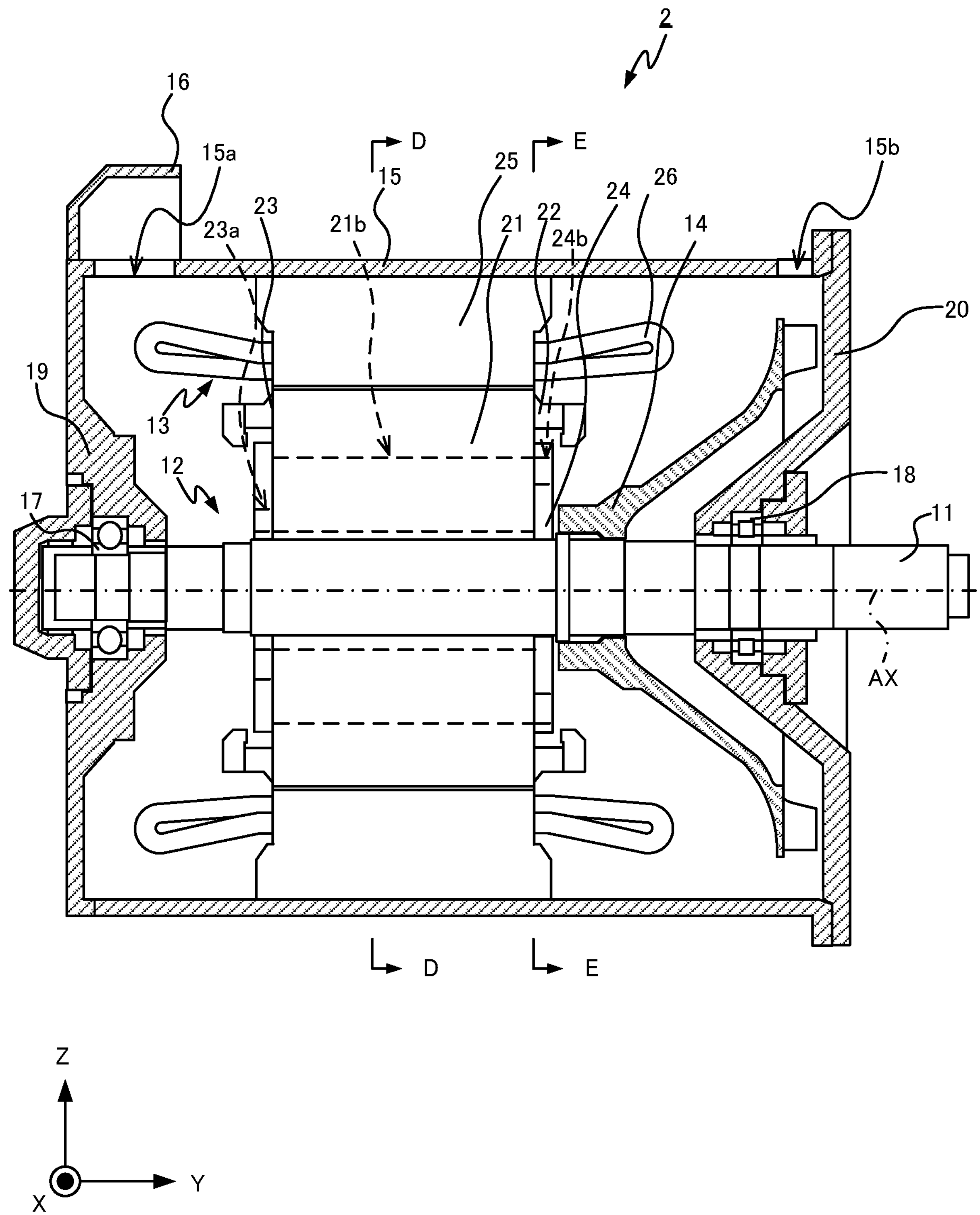


FIG.7



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16
15a
D
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15b
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23a
21b
21
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FIG.8

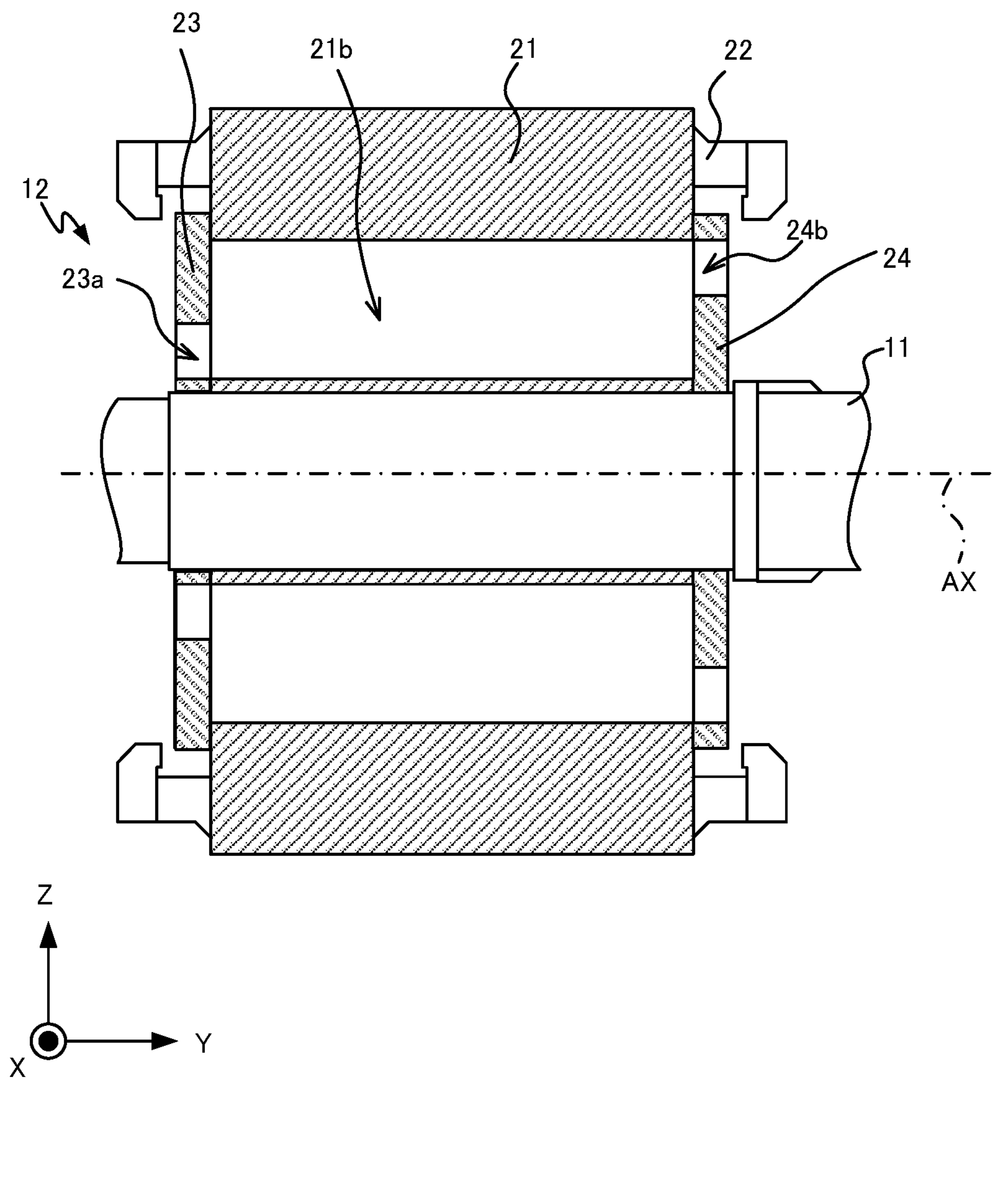

FIG.9

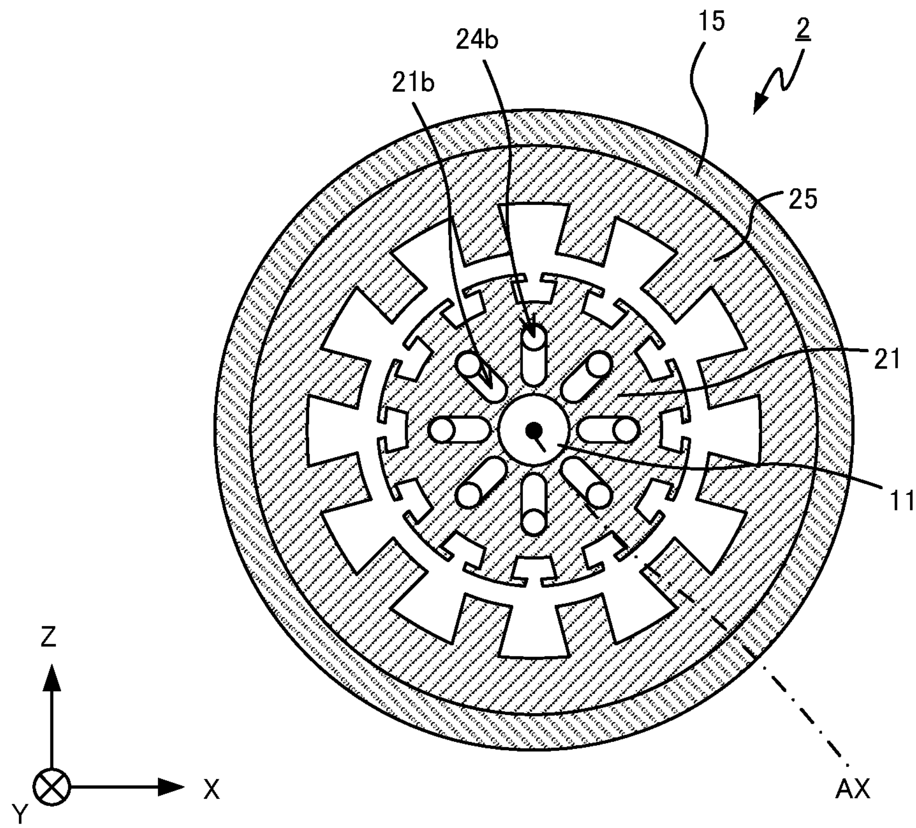

FIG.10

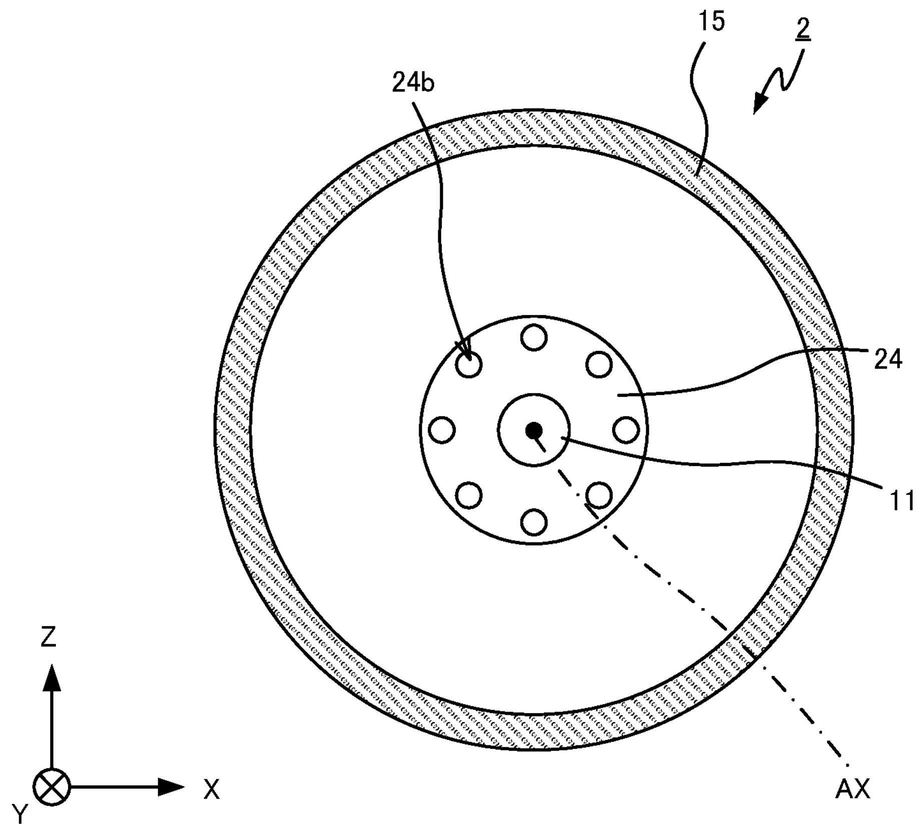

FIG.11

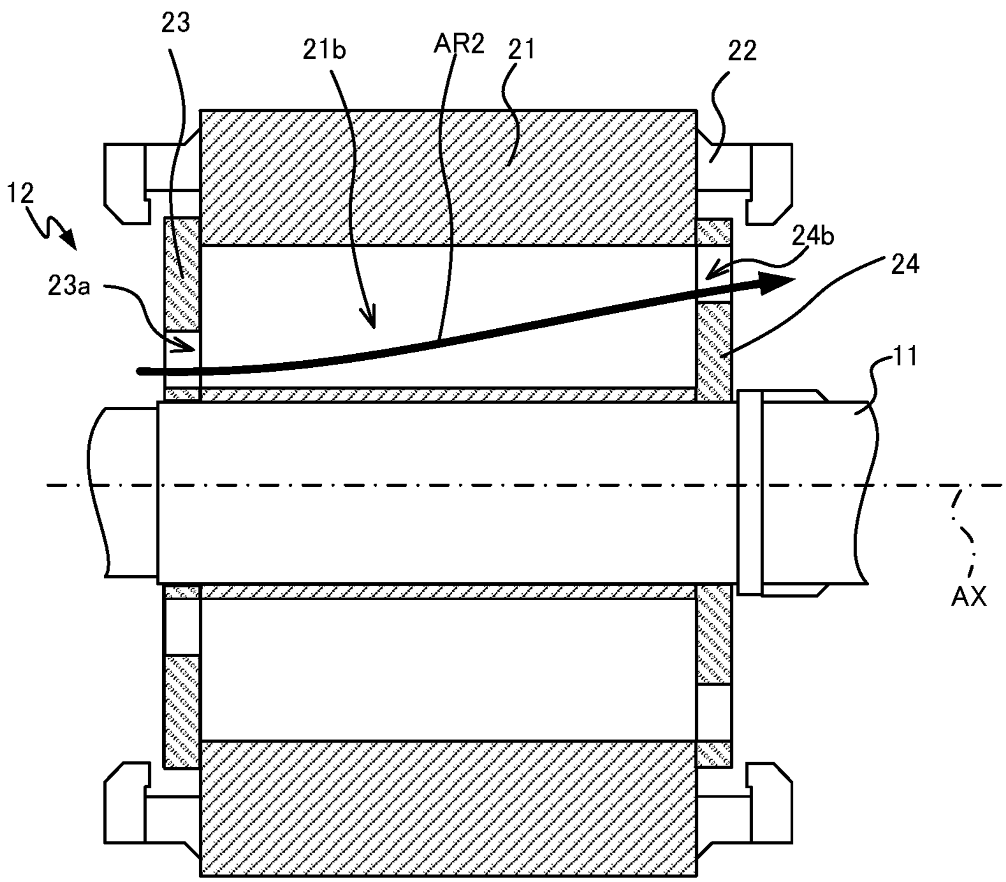

FIG.12

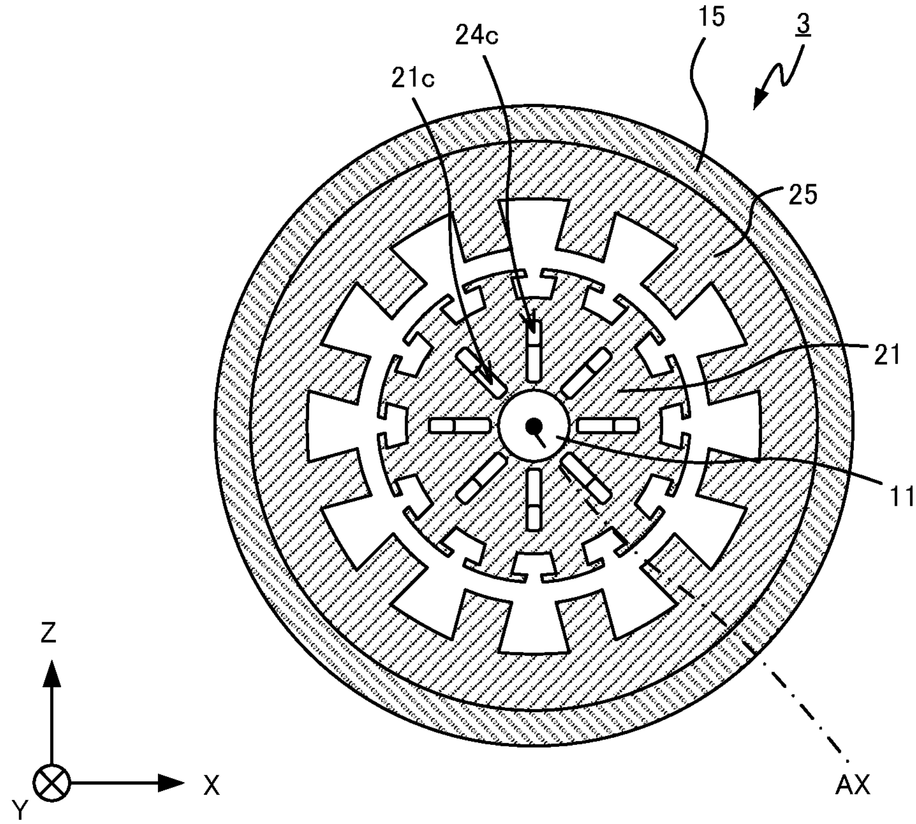

FIG.13

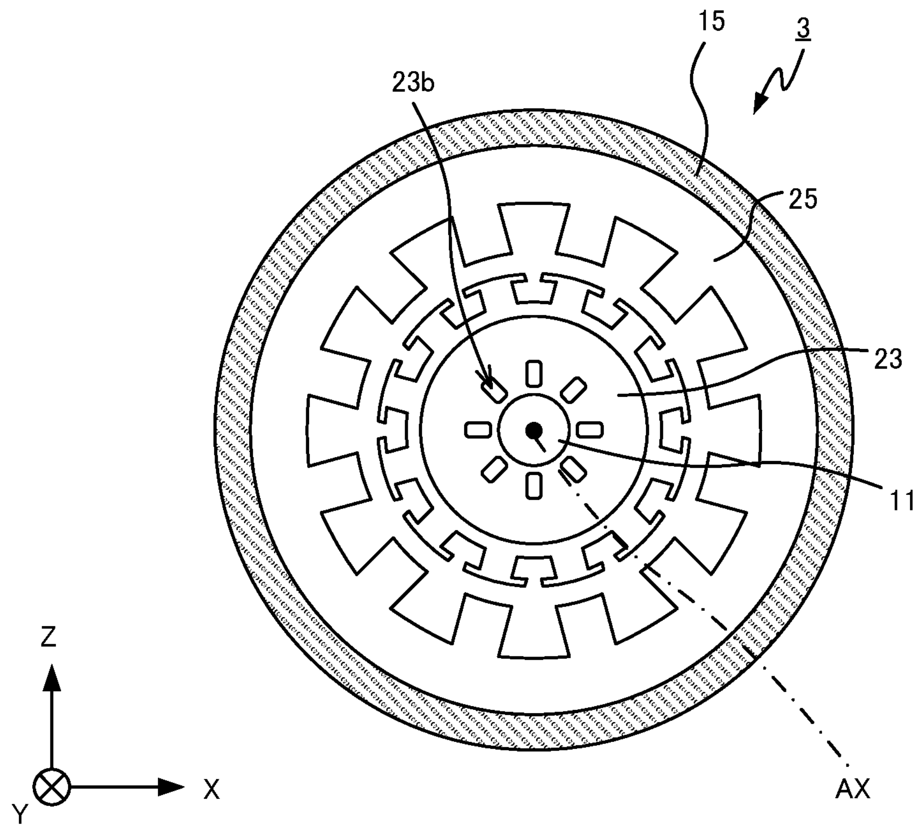

FIG.14

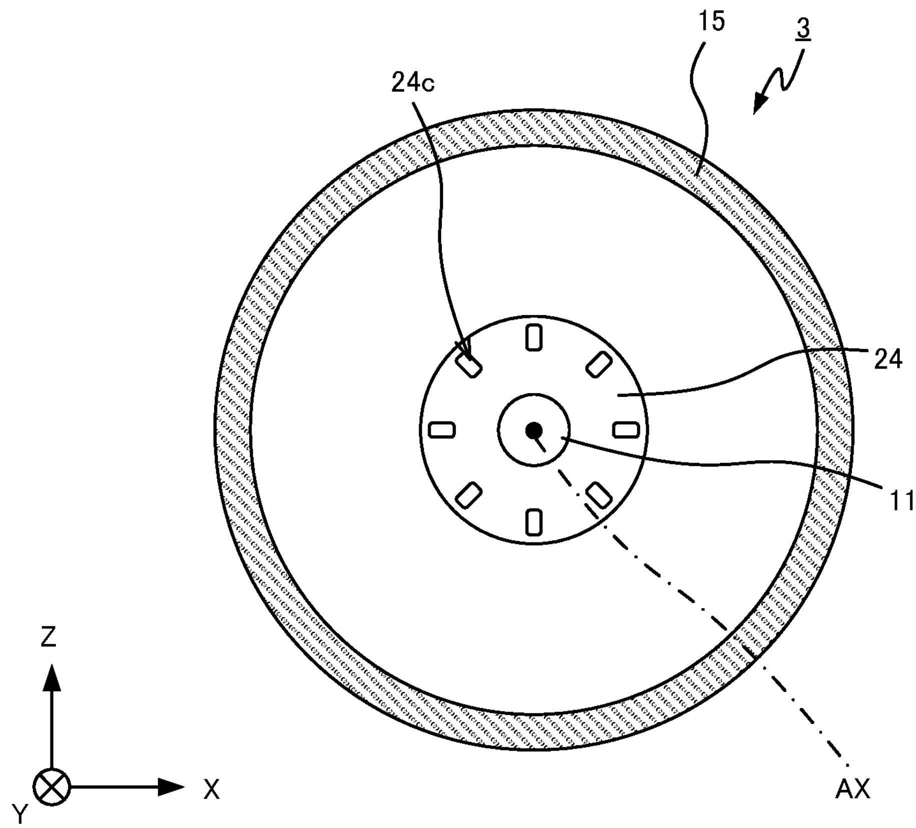

FIG.15

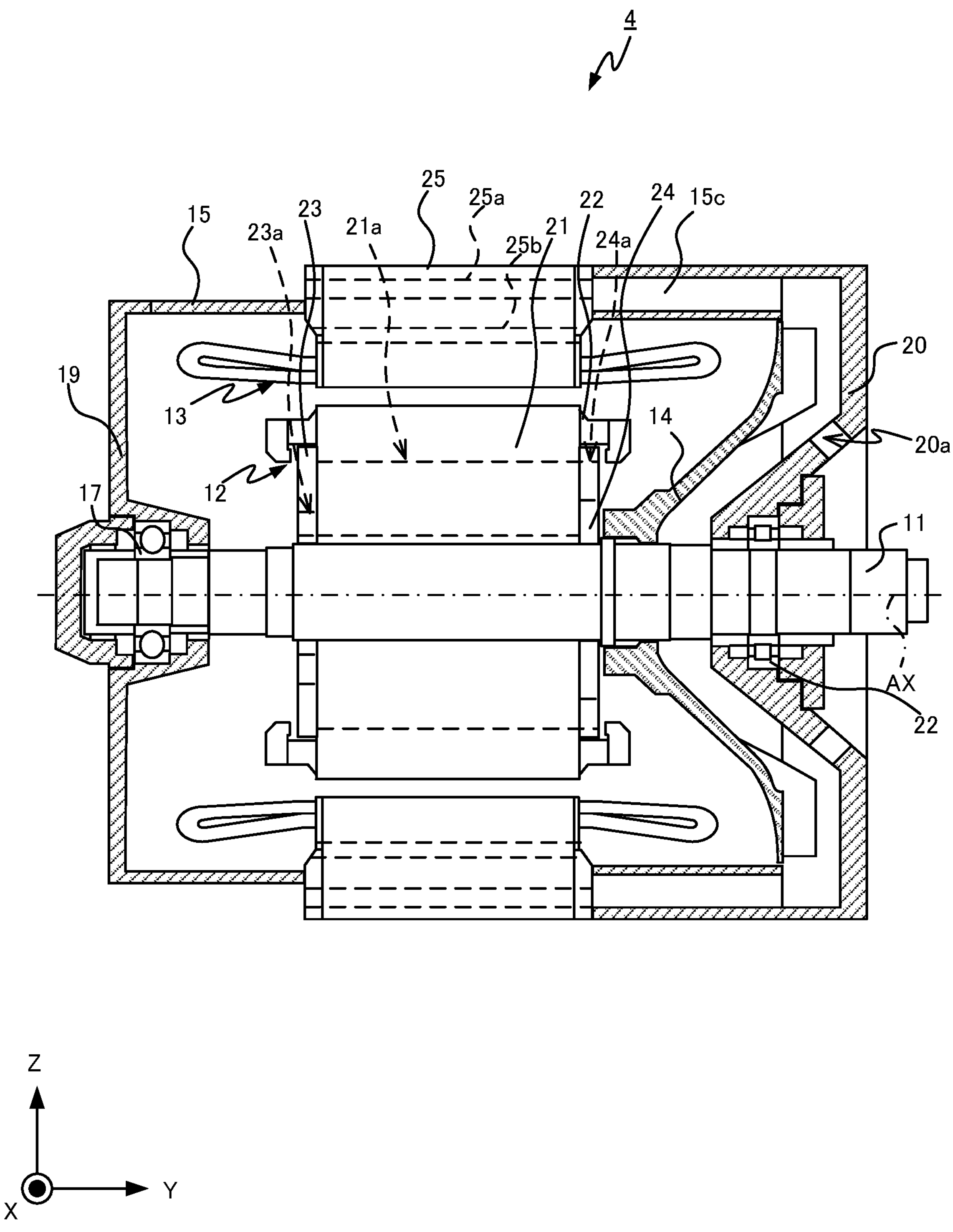
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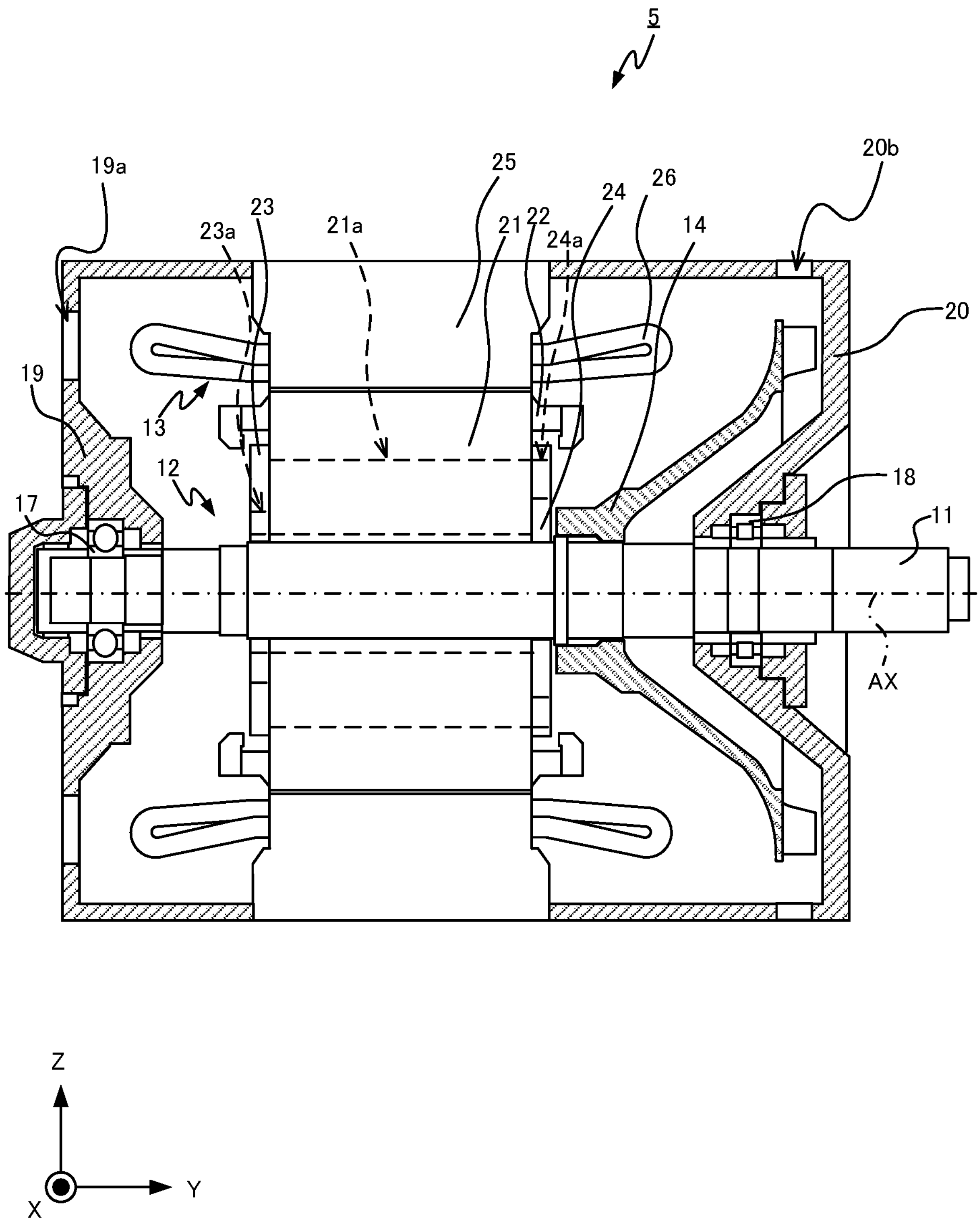
FIG.16



ELECTRIC MOTOR

TECHNICAL FIELD

The present disclosure relates to an electric motor.

BACKGROUND ART

Some electric motors each include a fan fixed to a shaft and rotatable integrally with the shaft, in order to cool components of the electric motor. In this electric motor, the rotation of the fan causes air outside the electric motor to enter the electric motor and flow through air passages of a rotor core and a gap between a stator core and the rotor core, for example. This air flow cools components of the electric motor, such as a stator core, a stator conductor, a rotor core, and a rotor conductor. A typical example of this type of electric motor is disclosed in Patent Literature 1. In the electric motor disclosed in Patent Literature 1, the air introduced into the electric motor by means of rotation of a fan flows through openings of core retainers and wind holes of a rotor core and then exits the electric motor. This configuration can cool a stator core, a stator conductor, the rotor core, and a rotor conductor.

CITATION LIST

Patent Literature

Patent Literature 1: Unexamined Japanese Utility Model Application Publication No. H6-48355

SUMMARY OF INVENTION

Technical Problem

The openings and the wind hole of the electric motor disclosed in Patent Literature 1 extend in a direction intersecting the rotation axis. The rotor core including the wind hole having this shape requires a complicated fabrication process that involves bonding a large number of thin metal plates in which through holes having the identical shape are disposed at mutually different positions, or boring the rotor core with a machine.

An objective of the present disclosure, which has been accomplished in view of the above situations, is to provide an electric motor that has high cooling efficiency and can be fabricated by a simple process.

Solution to Problem

In order to achieve the above objective, an electric motor according to an aspect of the present disclosure includes a shaft, a rotor, a stator, a first retaining member, and a second retaining member. The shaft is supported rotatably around a rotation axis. The rotor is located radially outward from the shaft and rotatable integrally with the shaft. The stator is radially opposing the rotor to define a space. The first retaining member and the second retaining member hold the rotor therebetween in the extending direction of the rotation axis. The rotor includes a stack of thin plates arranged in the extending direction of the rotation axis and including through holes having the identical shape. The through holes of the individual thin plates included in the stack define air passages extending through the rotor in the extending direction of the rotation axis. The first retaining member includes first through holes in communication with the upstream ends of the air passages. The second retaining member includes second through holes in communication with the downstream ends of the air passages. The radially outermost portion of the wall of each of the first through holes is located radially inward from the radially outermost portion of the wall of each of the second through holes.

Advantageous Effects of Invention

The rotor included in the electric motor according to an aspect of the present disclosure includes the air passages extending through the rotor in the extending direction of the rotation axis, and the first retaining member and the second retaining member holding the rotor therebetween in the extending direction of the rotation axis includes the first through holes and the second through holes, respectively. The radially outermost portion of the wall of each of the first through holes is located radially inward from the radially outermost portion of the wall of each of the second through holes. Accordingly, the air at the openings of the second through holes has a static pressure lower than that of the air at the openings of the first through holes during rotation of the rotor. This configuration facilitates the air to flow through the first through holes into the air passages, thereby improving the cooling efficiency. The electric motor, of which the rotor is made of a stack of thin plates including through holes having the identical shape, can be fabricated by a simple process.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a cross-sectional view of an electric motor according to Embodiment 1;

FIG. **2** is a partial cross-sectional view of the electric motor according to Embodiment 1;

FIG. **3** is a cross-sectional view of the electric motor according to Embodiment 1 taken along the line A-A of FIG. **1**;

FIG. **4** is a cross-sectional view of the electric motor according to Embodiment 1 taken along the line B-B of FIG. **1**;

FIG. **5** is a cross-sectional view of the electric motor according to Embodiment 1 taken along the line C-C of FIG. **1**;

FIG. **6** illustrates a flow of air inside a rotor included in the electric motor according to Embodiment 1;

FIG. **7** is a cross-sectional view of an electric motor according to Embodiment 2;

FIG. **8** is a partial cross-sectional view of the electric motor according to Embodiment 2;

FIG. **9** is a cross-sectional view of the electric motor according to Embodiment 2 taken along the line D-D of FIG. **7**;

FIG. **10** is a cross-sectional view of the electric motor according to Embodiment 2 taken along the line E-E of FIG. **7**;

FIG. **11** illustrates a flow of air inside a rotor included in the electric motor according to Embodiment 2;

FIG. **12** is a cross-sectional view of an electric motor according to a first modification of the embodiments;

FIG. **13** is another cross-sectional view of the electric motor according to the first modification of the embodiments;

FIG. **14** is a cross-sectional view of an electric motor according to a second modification of the embodiments;

FIG. **15** is a cross-sectional view of an electric motor according to a third modification of the embodiments; and

FIG. **16** is a cross-sectional view of an electric motor according to a fourth modification of the embodiments.

DESCRIPTION OF EMBODIMENTS

An electric motor according to embodiments of the present disclosure is described in detail below with reference to the accompanying drawings. In the drawings, the components identical or corresponding to each other are provided with the same reference symbol.

Embodiment 1

An electric motor **1** according to Embodiment 1 is described below with reference to FIGS. **1** to **6**, focusing on an exemplary open-type self-cooled electric motor for driving a railway vehicle. The open-type self-cooled electric motor introduces air from the outside into the electric motor by means of rotation of a fan and thus cools components of the electric motor. In FIGS. **1** to **6**, the Z axis corresponds to the vertical direction, the Y axis is parallel to a rotation axis AX of a shaft **11** of the electric motor **1**, and the X axis is orthogonal to the Y and Z axes. FIG. **1** illustrates the rotation axis AX with a dashed and single-dotted line. The air entering the electric motor **1** flows through first through holes **23***a* of a first retaining member **23**, air passages **21***a* of a rotor core **21** included in a rotor **12**, and second through holes **24***a* of a second retaining member **24** in sequence and then exits the electric motor **1**. This air flow inside the rotor **12** cools the rotor **12**, which is a component of the electric motor **1**. The second through holes **24***a* of the second retaining member **24** are located radially outward from the first through holes **23***a* of the first retaining member **23**. This arrangement facilitates air to enter the air passages **21***a*, leading to high cooling efficiency.

The electric motor **1** illustrated in FIG. **1** includes the shaft **11** supported rotatably around the rotation axis AX, the rotor **12** located radially outward from the shaft **11** and rotatable integrally with the shaft **11**, a stator **13** radially opposing the rotor **12**, a fan **14** rotatable integrally with the shaft **11**, and the first retaining member **23** and the second retaining member **24** to hold the rotor **12** therebetween in the extending direction of the rotation axis AX.

The electric motor **1** further includes a frame **15** to accommodate the shaft **11**, the rotor **12**, the stator **13**, the fan **14**, the first retaining member **23**, and the second retaining member **24**. The frame **15** includes, in a vertically upper portion thereof, an inlet hole **15***a* at one end in the Y-axis direction to introduce air from the outside, and an outlet hole **15***b* at the other end in the Y-axis direction to discharge the introduced air. The electric motor **1** also includes a cover **16** to prevent contaminants, such as dust and water drops, from entering the electric motor **1** via the inlet hole **15***a*. The electric motor **1** further includes bearings **17** and **18** to support the shaft **11** rotatably, a first bracket **19** to hold the bearing **17**, and a second bracket **20** to hold the bearing **18**.

The individual components of the electric motor **1** are described in more detail below.

The end of the shaft **11** adjacent to the second bracket **20** is coupled to an axle of the railway vehicle via a joint and gears, which are not illustrated. The rotation of the shaft **11** provides power to the railway vehicle.

The rotor **12** includes the rotor core **21** fixed to the shaft **11**, and rotor conductors **22** disposed in a groove on the outer peripheral surface of the rotor core **21**. Since the rotor core **21** is fixed to the shaft **11**, the rotor **12** including the rotor core **21** and the rotor conductor **22** rotates integrally with the shaft **11**.

The rotor core **21** is a stack of thin plates, which are arranged in the extending direction of the rotation axis AX and include through holes having the identical shape. In detail, the rotor core **21** is a stack of thin silicon steel plates, which are arranged in the extending direction of the rotation axis AX and include through holes having the identical shape. As illustrated in FIG. **1** and FIG. **2**, which is a partial enlarged view of FIG. **1**, the rotor core **21** made of such a stack includes the air passages **21***a* extending through the rotor core **21** in the extending direction of the rotation axis AX. In Embodiment 1, the ends of the air passages **21***a* that face the first retaining member **23** correspond to the upstream ends, and the ends of the air passages **21***a* that face the second retaining member **24** correspond to the downstream ends. Each of the air passages **21***a* has cross sections orthogonal to the penetration direction that have a constant shape from the upstream end to the downstream end. As illustrated in FIG. **3**, which is a cross-sectional view taken along the line A-A of FIG. **1**, the cross section of the air passage **21***a* orthogonal to the penetration direction preferably has a shape defined by connecting, with straight lines, the outer edges of a first circle and a second circle located radially outward from the first circle and having a larger diameter than that of the first circle. The major axis of the shape defined by connecting, with straight lines, the outer edges of the first circle and the second circle preferably radially extends. The sizes of the first circle and the second circle can be determined depending on a required cooling capacity of the electric motor **1** and a required strength of the rotor core **21** to receive torque.

As illustrated in FIG. **2**, the air passages **21***a* are in communication with both of the first through holes **23***a* of the first retaining member **23** and the second through holes **24***a* of the second retaining member **24**. The air passages **21***a* preferably have a radial length larger than that of the first through holes **23***a*. The air passages **21***a* preferably have a radial length larger than that of the second through holes **24***a*.

As illustrated in FIG. **1**, the stator **13** includes a stator core **25** fixed to the frame **15**, and stator conductors **26** disposed in a groove on the stator core **25**. The stator core **25** radially opposes the rotor core **21** to define a space therebetween.

The fan **14** is fixed to the shaft **11** at a position adjacent to the second retaining member **24** and rotates integrally with the shaft **11**. In Embodiment 1, the fan **14** is fixed to the shaft **11** such that the main surface faces the second bracket **20**.

The frame **15** has a hollow cylindrical shape. The openings of the frame **15** at both ends in the extending direction of the rotation axis AX are closed by the first bracket **19** and the second bracket **20**. The frame **15** includes, in vertically upper portions of the outer peripheral surface, the inlet hole **15***a* to introduce air from the outside, and the outlet hole **15***b* to discharge the air introduced via the inlet hole **15***a* and flowing through the air passages **21***a*.

The cover **16** is disposed over the inlet hole **15***a* of the frame **15**.

The bearings **17** and **18** support the shaft **11** such that the shaft **11** is rotatable.

The first bracket **19** holds the bearing **17** and closes the opening of the hollow cylindrical frame **15** at one end.

The second bracket **20** holds the bearing **18** and closes the opening of the hollow cylindrical frame **15** at the other end.

The first retaining member **23** and the second retaining member **24** hold the rotor core **21** therebetween in the extending direction of the rotation axis AX. The first retaining member **23** and the second retaining member **24** are fixed to the rotor core **21** and rotate integrally with the rotor core **21**. As illustrated in FIG. **1** and FIG. **4**, which is a cross-sectional view taken along the line B-B of FIG. **1**, the first retaining member **23** includes the first through holes **23***a* in communication with the upstream ends of the air passages **21***a* of the rotor core **21**. As illustrated in FIG. **1** and FIG. **5**, which is a cross-sectional view taken along the line C-C of FIG. **1**, the second retaining member **24** includes the second through holes **24***a* in communication with the downstream ends of the air passages **21***a* of the rotor core **21**. The radially outermost portion of the wall of each of the first through holes **23***a* is located radially inward from the radially outermost portion of the wall of each of the second through holes **24***a*.

In the above-described example in which the cross sections of the air passages **21***a* orthogonal to the penetration direction have a shape defined by connecting, with straight lines, the outer edges of the first circle and the second circle located radially outward from the first circle and having a larger diameter than that of the first circle, the first through holes **23***a* preferably have cross sections orthogonal to the penetration direction having the shape identical to that of the first circle. In this case, the radially inner walls of the first through holes **23***a* smoothly continue to the radially inner walls of the air passages **21***a*.

In the above-described example in which the cross sections of the air passages **21***a* orthogonal to the penetration direction have a shape defined by connecting, with straight lines, the outer edges of the first circle and the second circle located radially outward from the first circle and having a larger diameter than that of the first circle, the second through holes **24***a* preferably have cross sections orthogonal to the penetration direction having the shape identical to that of the second circle. In this case, the radially outer walls of the second through holes **24***a* smoothly continue to the radially outer walls of the air passages **21***a*.

When the electric motor **1** having the above-described configuration is energized and causes the rotor core **21** and the shaft **11** to rotate integrally with each other, the fan **14** rotates along with the shaft **11**, so that air outside the electric motor **1** enters the electric motor **1** via the inlet hole **15***a* illustrated in FIG. **1**. The air introduced via the inlet hole **15***a* then arrives at the first through holes **23***a*. FIG. **6** illustrates a flow of air through the first retaining member **23**, the rotor core **21**, and the second retaining member **24** with an arrow AR**1**. As illustrated with the arrow AR**1**, the air arriving at the first through holes **23***a* flows through the first through holes **23***a*, the air passages **21***a*, and then the second through holes **24***a*. The air flowing through the second through holes **24***a* exits the electric motor **1** via the outlet hole **15***b* illustrated in FIG. **1**. The above-described air flows inside the electric motor **1** cool the electric motor **1**.

As described above, the first through holes **23***a* are located radially inward from the second through holes **24***a* in the electric motor **1** according to Embodiment 1. Accordingly, the air at the first through holes **23***a* has a static pressure higher than that of the air at the second through holes **24***a* during rotation of the rotor **12**. This configuration facilitates the air to flow through the first through holes **23***a*, the air passages **21***a*, and the second through holes **24***a*, thereby improving the cooling efficiency. The electric motor **1**, of which the rotor core **21** is made of a stack of thin plates

including through holes having the identical shape, can be fabricated by a simple process.

In the case where the cross sections of the air passage **21***a* orthogonal to the penetration direction have a shape defined by connecting, with straight lines, the outer edges of the first circle and the second circle located radially outward from the first circle and having a larger diameter than that of the first circle, a larger amount of air can be sent to portions of the air passages **21***a* adjacent to the rotor conductor **22** that emits heat, leading to more efficient cooling of the electric motor **1**.

Embodiment 2

The air passages **21***a*, the first through holes **23***a*, and the second through holes **24***a* may have any shape other than that of the above-described example, provided that the air introduced via the inlet hole **15***a* is guided to the outlet hole **15***b*. The description of Embodiment 2 is directed to an electric motor **2** including air passages **21***b* and second through holes **24***b* having different shapes from those in Embodiment 1.

The structure of the electric motor **2** illustrated in FIG. **7** is generally identical to that in Embodiment 1. The rotor core **21** of the electric motor **2** includes the air passages **21***b* extending through the rotor core **21**, as illustrated in FIG. **8**, which is a partial enlarged view of FIG. **7**. In Embodiment 2, the ends of the air passages **21***b* that face the first retaining member **23** correspond to the upstream ends, and the ends of the air passages **21***b* that face the second retaining member **24** correspond to the downstream ends. The upstream ends of the air passages **21***b* are in communication with the first through holes **23***a* of the first retaining member **23**. The downstream ends of the air passages **21***b* are in communication with the second through holes **24***b* of the second retaining member **24**. Each of the air passages **21***b* has cross sections orthogonal to the penetration direction that have a constant shape from the upstream end to the downstream end. As illustrated in FIG. **9**, which is a cross-sectional view taken along the line D-D of FIG. **7**, the cross section of the air passage **21***b* orthogonal to the penetration direction has a shape defined by connecting, with straight lines, the outer edges of a first circle and a second circle located radially outward from the first circle and having the same diameter as that of the first circle.

As illustrated in FIG. **8** and FIG. **10**, which is a cross-sectional view taken along the line E-E of FIG. **7**, the second retaining member **24** includes the second through holes **24***b* in communication with the downstream ends of the air passages **21***a* of the rotor core **21**. The radially outermost portion of the wall of each of the first through holes **23***a* is located radially inward from the radially outermost portion of the wall of each of the second through holes **24***b*, as in Embodiment 1. The second through holes **24***b* have cross sections orthogonal to the penetration direction having the same shape as that of the above-mentioned second circle. The radially outer walls of the second through holes **24***b* smoothly continue to the radially outer walls of the air passages **21***b*.

When the electric motor **2** is energized and causes the rotor core **21** and the shaft **11** to rotate integrally with each other, the fan **14** rotates along with the shaft **11**, so that the air outside the electric motor **2** enters the electric motor **2** via the inlet hole **15***a* illustrated in FIG. **7**. The air entering via the inlet hole **15***a* then arrives at the first through holes **23***a*. FIG. **11** illustrates a flow of air through the first retaining member **23**, the rotor core **21**, and the second retaining member **24** with an arrow AR**2**. As illustrated with the arrow AR**2**, the air arriving at the first through hole **23***a* flows through the first through hole **23***a*, the air passage **21***b*, and then the second through hole **24***b*. The air flowing through the second through hole **24***b* exits the electric motor **2** via the outlet hole **15***b* illustrated in FIG. **7**. The above-described air flows inside the electric motor **2** cool the electric motor **2**.

As described above, the first through holes **23***a* are located radially inward from the second through holes **24***b* in the electric motor **2** according to Embodiment 2. Accordingly, the air at the first through holes **23***a* has a static pressure higher than that of the air at the second through holes **24***b* during rotation of the rotor **12**. This configuration facilitates the air to flow through the first through holes **23***a*, the air passages **21***b*, and the second through holes **24***b*, thereby improving the cooling efficiency. The electric motor **2**, of which the rotor core **21** is made of a stack of thin plates including through holes having the identical shape, can be fabricated by a simple process.

The above-described embodiments are not to be construed as limiting the present disclosure.

Provided that the rotor core **21** is a stack of thin plates including through holes having the identical shape, these through holes may have any shape. For example, an electric motor **3** may include the rotor core **21** made of a stack of thin plates including through holes having a rectangular shape with rounded corners. In detail, air passages **21***c* illustrated in FIG. **12** have cross sections orthogonal to the penetration direction having a rectangular shape with rounded corners. For another example, the air passages **21***a*, **21***b*, and **21***c* may have cross sections orthogonal to the penetration direction having a wing or elliptical shape.

The number and positions of the air passages **21***a*, **21***b*, and **21***c* are not necessarily the number and positions in the above-described examples, and can be appropriately determined depending on parameters, such as a required cooling capacity of the electric motors **1** to **3** and a required strength of the rotor core **21** to receive torque.

The cross sections of the first through holes **23***a* orthogonal to the penetration direction may have any shape, provided that the first through holes **23***a* are located radially inward from the second through holes **24***a* or **24***b* and able to guide the air introduced via the inlet hole **15***a* to the air passages **21***a* or **21***b*. For example, the first retaining member **23** of the electric motor **3** may include first through holes **23***b* illustrated in FIG. **13**. These first through holes **23***b* have cross sections orthogonal to the penetration direction having a rectangular shape with rounded corners. For another example, the first through holes **23***a* and **23***b* may have cross sections orthogonal to the penetration direction having a wing or elliptical shape.

The first through holes **23***a* and **23***b* may have a circumferential width larger than that of the air passages **21***a* and **21***b*. The penetration direction of first through holes **23***a* and **23***b* is not limited to the extending direction of the rotation axis AX.

The cross sections of the second through holes **24***a* and **24***b* orthogonal to the penetration direction may have any shape, provided that the second through holes **24***a* or **24***b* are located radially outward from the first through holes **23***a* or **23***b* and able to discharge the air flowing from the air passages **21***a*, **21***b*, or **21***c*. For example, the second retaining member **24** of the electric motor **3** may include second through holes **24***c* illustrated in FIG. **14**. These second through holes **24***c* have cross sections orthogonal to the penetration direction having a rectangular shape with rounded corners. For another example, the second through holes **24***a*, **24***b*, and **24***c* may have cross sections orthogonal to the penetration direction having a wing or elliptical shape.

The second through holes **24***a*, **24***b*, and **24***c* may have a circumferential width larger than that of the air passages **21***a*, **21***b*, and **21***c*. The second through holes **24***a*, **24***b*, and **24***c* may have the same radial length as that of the air passages **21***a*, **21***b*, and **21***c*. The second through holes **24***a*, **24***b*, and **24***c* may extend in a direction other than the extending direction of the rotation axis AX.

The first retaining member **23** and the rotor core **21** may be provided with an end plate therebetween. In this case, the end plate between the first retaining member **23** and the rotor core **21** includes through holes having the same shape as that of the air passages **21***a*, **21***b*, or **21***c*. Also, the second retaining member **24** and the rotor core **21** may be provided with an end plate therebetween. In this case, the end plate between the second retaining member **24** and the rotor core **21** includes through holes having the same shape as that of the air passages **21***a*, **21***b*, or **21***c*.

The present disclosure can also be applied to a closed-type electric motor other than the open-type self-cooled electric motor. FIG. **15** illustrates a closed-type electric motor **4**. As in the electric motor **1** according to Embodiment 1, the rotor core **21** includes the air passages **21***a*, the first retaining member **23** includes the first through holes **23***a*, and the second retaining member **24** includes the second through holes **24***a*. Unlike Embodiment 1, the frame **15** of the electric motor **4** includes air passages **15***c*, the second bracket **20** includes inlet holes **20***a*, and the stator core **25** includes outer air passages **25***a* and inner air passages **25***b*. The inner air passages **25***b* are located radially inward from the outer air passages **25***a*.

When the electric motor **4** is energized, the rotation of the fan **14** causes air to enter the electric motor **4** via the inlet holes **20***a*. The air entering the electric motor **4** flows through the air passages **15***c* and the outer air passages **25***a* and then exits the electric motor **4**. Inside the electric motor **4**, the air arriving at the inner air passages **25***b* by means of rotation of the fan **14** flows through the inner air passages **25***b* and then arrives at the first through holes **23***a*. The air arriving at the first through holes **23***a* flows through the first through holes **23***a*, the air passages **21***a*, and then the second through holes **24***a*, as in Embodiment 1. The air flowing through the second through holes **24***a* arrives at the inner air passages **25***b* again and circulates as described above. The flows of air involving entering the electric motor **4**, flowing through the stator core **25**, and exiting the electric motor **4** and the circulation of the air inside the electric motor **4** cool components of the electric motor **4**.

The present disclosure can also be applied to a frameless electric motor. An electric motor **5** illustrated in FIG. **16** lacks the frame **15**, unlike the electric motors **1** to **4**. The first bracket **19** of the electric motor **5** includes inlet holes **19***a* to introduce air from the outside, and the second bracket **20** includes outlet holes **20***b* to discharge the air. As in the electric motor **1** according to Embodiment 1, in the electric motor **5**, the rotor core **21** includes the air passages **21***a*, the first retaining member **23** includes the first through holes **23***a*, and the second retaining member **24** includes the second through holes **24***a*.

When the electric motor **5** is energized, the rotation of the fan **14** causes air to enter the electric motor **5** via the inlet holes **19***a*. The air entering the electric motor **5** arrives at the first through holes **23***a*. The air arriving at the first through holes **23***a* flows through the first through holes **23***a*, the air passages **21***a*, and then the second through holes **24***a*, as in Embodiment 1. The air flowing through the second through holes **24**_a_ exits the electric motor **5** via the outlet holes **20**_b_. The air entering the electric motor **5** and flowing inside the rotor **12** cools components of the electric motor **5**.

The foregoing describes some example embodiments for explanatory purposes. Although the foregoing discussion has presented specific embodiments, persons skilled in the art will recognize that changes may be made in form and detail without departing from the broader spirit and scope of the invention. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. This detailed description, therefore, is not to be taken in a limiting sense, and the scope of the invention is defined only by the included claims, along with the full range of equivalents to which such claims are entitled.

REFERENCE SIGNS LIST

1, **2**, **3**, **4**, **5** Electric motor
11 Shaft
12 Rotor
13 Stator
14 Fan
15 Frame
15_a_, **19**_a_, **20**_a_ Inlet hole
15_b_, **20**_b_ Outlet hole
15_c_ Air passage
16 Cover
17, **18** Bearing
19 First bracket
20 Second bracket
21 Rotor core
21_a_, **21**_b_, **21**_c_ Air passage
22 Rotor conductor
23 First retaining member
23_a_, **23**_b_ First through hole
24 Second retaining member
24_a_, **24**_b_, **24**_c_ Second through hole
25 Stator core
25_a_ Outer air passage
25_b_ Inner air passage
26 Stator conductor
AR**1**, AR**2** Arrow
AX Rotation axis

The invention claimed is:

1. An electric motor comprising:
a shaft supported rotatably around a rotation axis;
a rotor located radially outward from the shaft and rotatable integrally with the shaft;
a stator radially opposing the rotor to define a space therebetween; and
a first retaining member and a second retaining member to hold the rotor therebetween in an extending direction of the rotation axis, wherein
the rotor comprises a stack of thin plates arranged in the extending direction of the rotation axis and having a mutually-identical shape, the thin plates each including a through hole having an identical shape,
the through hole of the individual thin plate included in the stack defines an air passage extending through the rotor in the extending direction of the rotation axis,
the first retaining member includes a first through hole in communication with an upstream end of the air passage,
the second retaining member includes a second through hole in communication with a downstream end of the air passage,
a radially outermost portion of a wall of the first through hole is located radially inward from a radially innermost portion of a wall of the second through hole,
the second through hole is located radially outward from the first through hole,
a cross section of the air passage orthogonal to a penetration direction has a shape defined by connecting, with straight lines, outer edges of a first semicircle and a second semicircle, the second semicircle being located radially outward from the first semicircle,
a major axis of the shape defined by connecting, with straight lines, the outer edges of the first semicircle and the second semicircle extends radially,
a cross section of the first through hole orthogonal to a penetration direction has a shape overlapping a shape of the first semicircle,
a radially inner wall of the first through hole smoothly continues to a radially inner wall of the air passage,
a cross section of the second through hole orthogonal to a penetration direction has a shape overlapping a shape of the second semicircle, and
a radially outer wall of the second through hole smoothly continues to a radially outer wall of the air passage.

2. The electric motor according to claim **1**, wherein a radial length of the air passage is longer than a radial length of the first through hole.

3. The electric motor according to claim **2**, wherein a radial length of the air passage is longer than a radial length of the second through hole.

4. The electric motor according to claim **1**, wherein a radial length of the air passage is longer than a radial length of the second through hole.

5. The electric motor according to claim **1**, wherein a diameter of the second semicircle is larger than a diameter of the first semicircle.

6. The electric motor according to claim **1**, further comprising a fan fixed to the shaft at a position adjacent to the second retaining member and rotatable integrally with the shaft.

7. The electric motor according to claim **1**, further comprising a frame to accommodate the shaft, the rotor, the stator, the first retaining member, and the second retaining member, wherein
the frame includes:
an inlet hole to introduce air from an outside; and
an outlet hole to discharge the air to the outside, the air being introduced via the inlet hole and flowing through the first through hole, the air passage, and the second through hole.

8. The electric motor according to claim **1**, further comprising a pair of brackets to hold the stator therebetween in the extending direction of the rotation axis and accommodate the shaft, the rotor, the first retaining member, and the second retaining member.

9. The electric motor according to claim **8**, wherein
one of the pair of brackets includes an inlet hole to introduce air from an outside, and
the other of the pair of brackets includes an outlet hole to discharge the air to the outside, the air being introduced via the inlet hole and flowing through the first through hole, the air passage, and the second through hole.

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